Exemple d'utilisation du package levyBridge

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Préparation

Installation des packages depuis GitHub : `levyBridge` pour les ponts, `levysim` pour simuler des trajectoires et `levyfit` pour estimer les paramètres.

```
remotes::install_github("rytracetito/levy_packages", subdir = "levyBridge")
remotes::install_github("rytracetito/levy_packages", subdir = "levysim")
remotes::install_github("rytracetito/levy_packages", subdir = "levyfit")
```

Une trajectoire est observée à des instants irréguliers $0 = t_0 < \cdots < t_N$. Le pont stochastique tire, sur la grille régulière $t'_i = i\,t_N/N$, des valeurs qui suivent exactement la loi du processus sachant les observations (méthode `"Bridge"` de Grall-Maës, 2012). Sous le modèle, les accroissements reconstruits sont indépendants et de même loi : ce sont eux que l'on teste ensuite.

Paramètres utilisés :

Processus	Pont	Paramètres du processus	Paramètres du pont
Gamma	`gamma_bridge`	`a = 2, b = 3`	`a`
Wiener	`wiener_bridge`	`mu = 0.5, sigma = 1.5`	`sigma`
Inverse gaussien	`ig_bridge`	`mu = 1, eta = 3`	`eta`
Variance-gamma	`vg_bridge`	`theta = 0.5, sigma = 1.5, nu = 2`	`theta, sigma, nu`
Meixner	`meixner_bridge`	`a = 1, b = 0.3, d = 1.5, m = 0.2`	`a, d, m`

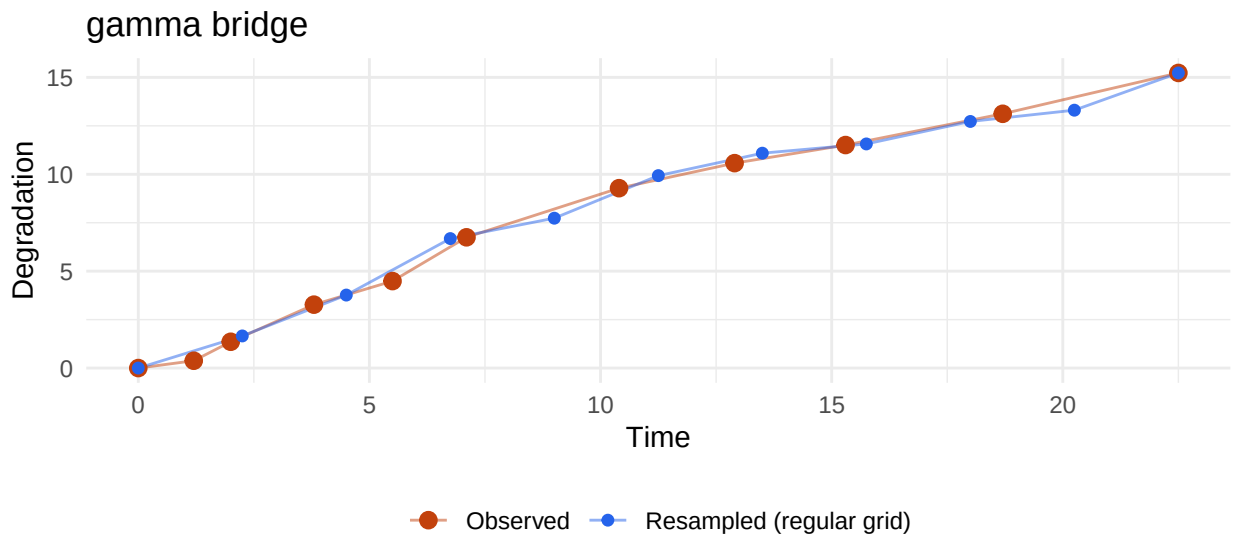
I - Une trajectoire

Une trajectoire observée à 11 instants irréguliers entre 0 et 22.5 ; la grille régulière est 0, 2.25, 4.5, . . . , 22.5.

```
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
```

Gamma

```
set.seed(1)
x <- sim_gamma(t, a = 2, b = 3)
r <- gamma_bridge(x, a = 2)
plot_bridge(r)
```

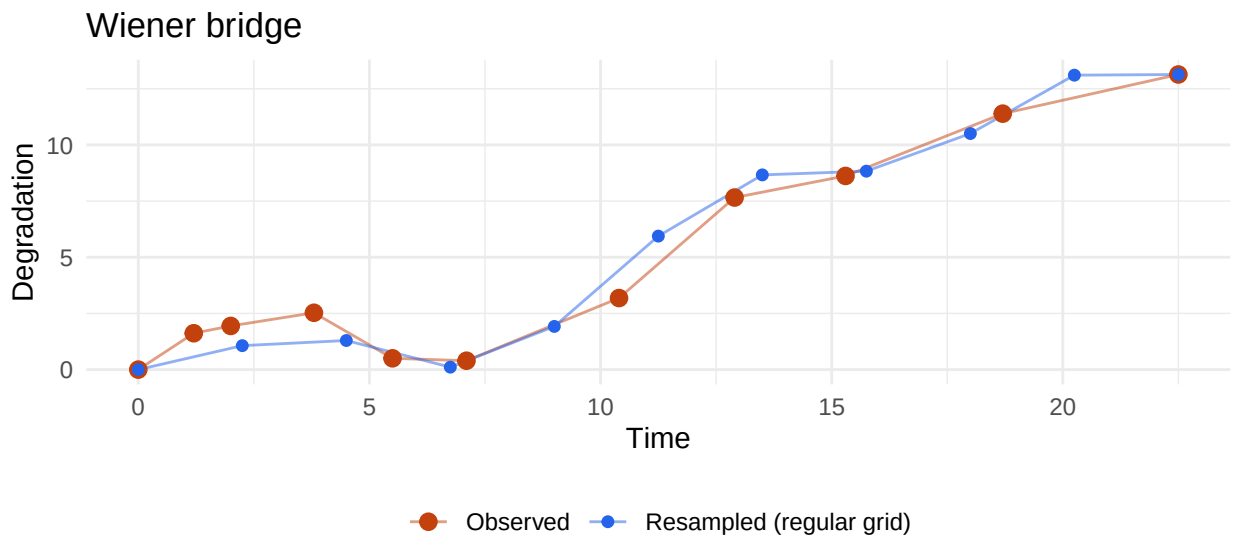



```
r[[1]]
```

```
##         t         X
## 1    0.00  0.000000
## 2    2.25  1.657691
## 3    4.50  3.766966
## 4    6.75  6.682118
## 5    9.00  7.734191
## 6   11.25  9.930251
## 7   13.50 11.088477
## 8   15.75 11.565691
## 9   18.00 12.725810
## 10 20.25 13.307300
## 11 22.50 15.229411
```

Wiener

```
x <- sim_wiener(t, mu = 0.5, sigma = 1.5)
r <- wiener_bridge(x, sigma = 1.5)
plot_bridge(r)
```

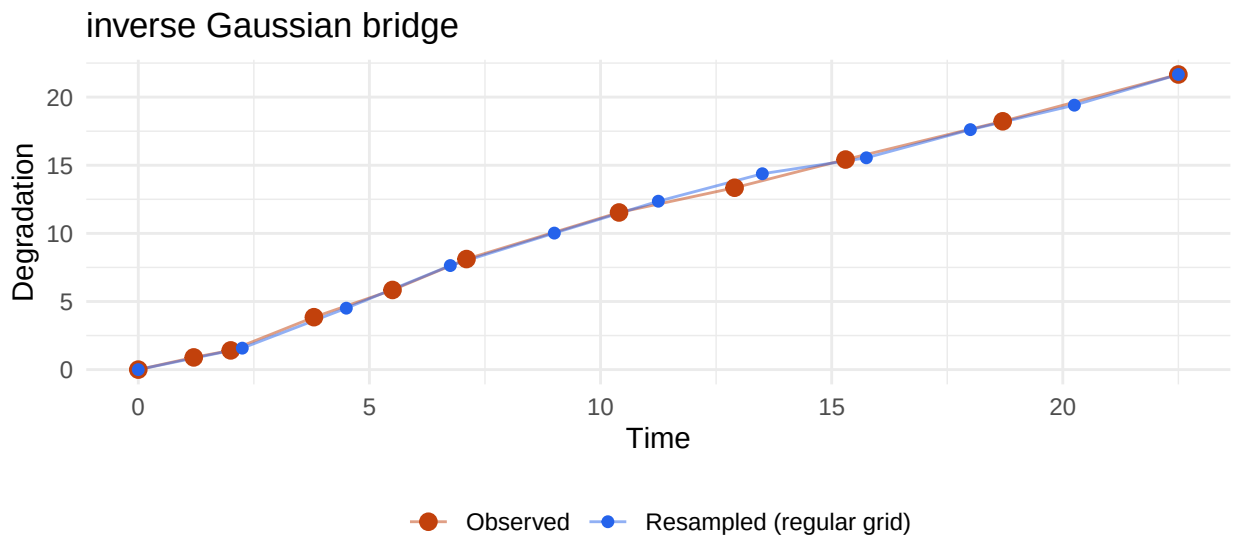



```
r[[1]]
```

```
##        t          X
## 1   0.00  0.0000000
## 2   2.25  1.0662340
## 3   4.50  1.2957781
## 4   6.75  0.1101619
## 5   9.00  1.9217942
## 6  11.25  5.9413083
## 7  13.50  8.6654258
## 8  15.75  8.8310626
## 9  18.00 10.5055605
## 10 20.25 13.1029615
## 11 22.50 13.1332709
```

Inverse gaussien

```
x <- sim_ig(t, mu = 1, eta = 3)
r <- ig_bridge(x, eta = 3)
plot_bridge(r)
```

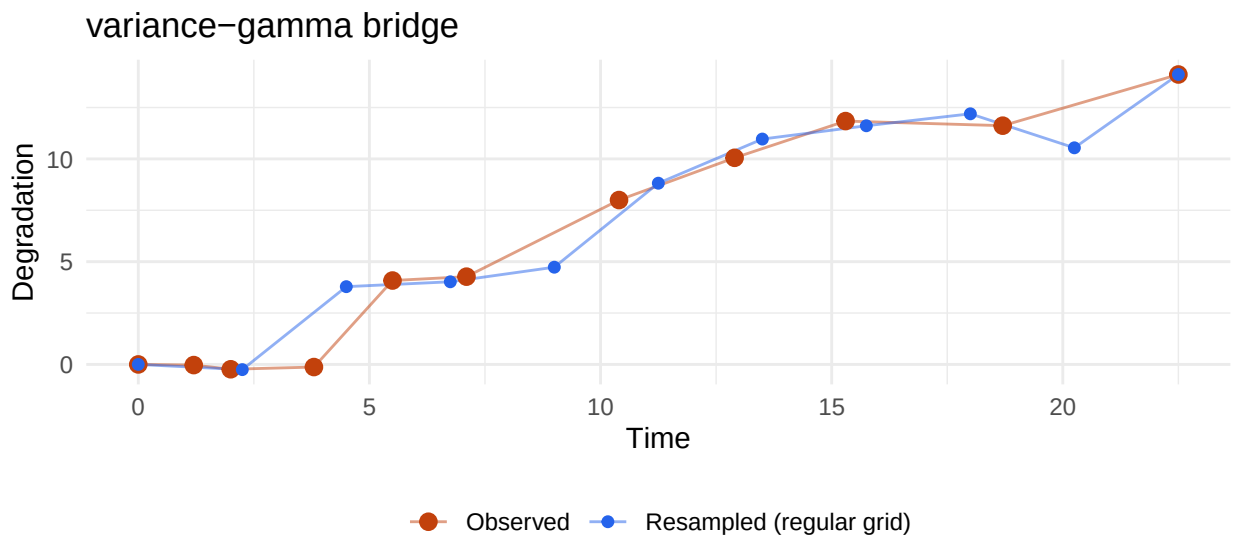



```
r[[1]]
```

```
##         t         X
## 1    0.00  0.000000
## 2    2.25  1.574758
## 3    4.50  4.508963
## 4    6.75  7.637565
## 5    9.00 10.019446
## 6   11.25 12.359964
## 7   13.50 14.376303
## 8   15.75 15.545750
## 9   18.00 17.613520
## 10 20.25 19.408542
## 11 22.50 21.660949
```

Variance-gamma

```
x <- sim_vg(t, theta = 0.5, sigma = 1.5, nu = 2)
r <- vg_bridge(x, theta = 0.5, sigma = 1.5, nu = 2)
plot_bridge(r)
```

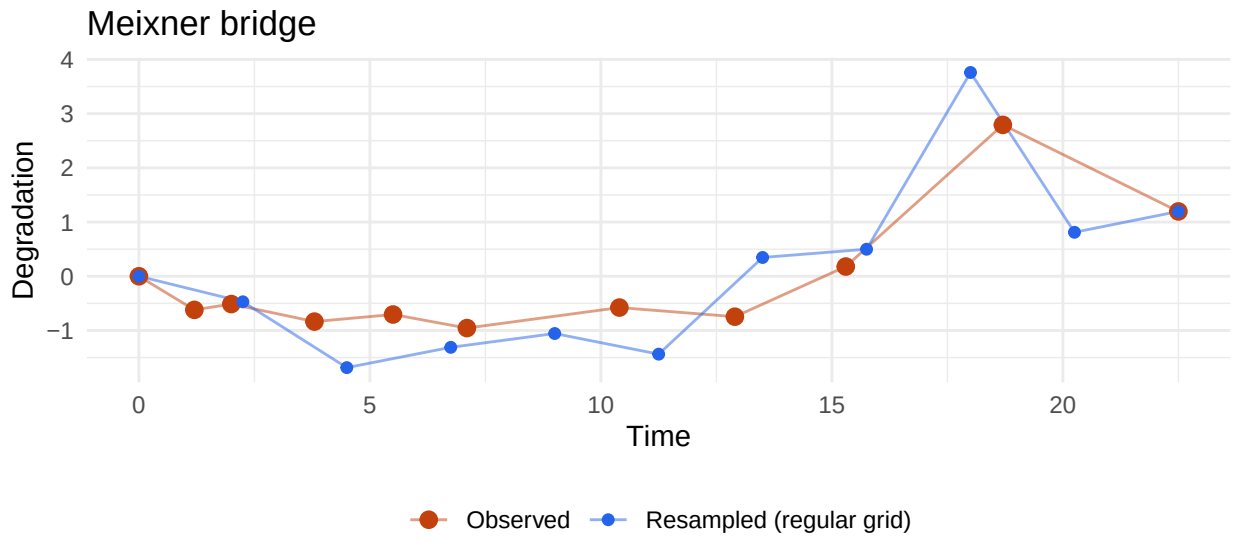



```
r[[1]]
```

```
##        t          X
## 1   0.00  0.0000000
## 2   2.25 -0.2521828
## 3   4.50  3.7840136
## 4   6.75  4.0215649
## 5   9.00  4.7306485
## 6  11.25  8.8172406
## 7  13.50 10.9650553
## 8  15.75 11.6120121
## 9  18.00 12.1930999
## 10 20.25 10.5394625
## 11 22.50 14.1063687
```

Meixner

```
x <- sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2)
r <- meixner_bridge(x, a = 1, d = 1.5, m = 0.2)
plot_bridge(r)
```




```
r[[1]]
```

```
##         t          X
## 1    0.00  0.0000000
## 2    2.25 -0.4711287
## 3    4.50 -1.6852487
## 4    6.75 -1.3110520
## 5    9.00 -1.0562761
## 6   11.25 -1.4370755
## 7   13.50  0.3466556
## 8   15.75  0.4986636
## 9   18.00  3.7591619
## 10 20.25  0.8111764
## 11 22.50  1.1960660
```

II - Plusieurs trajectoires

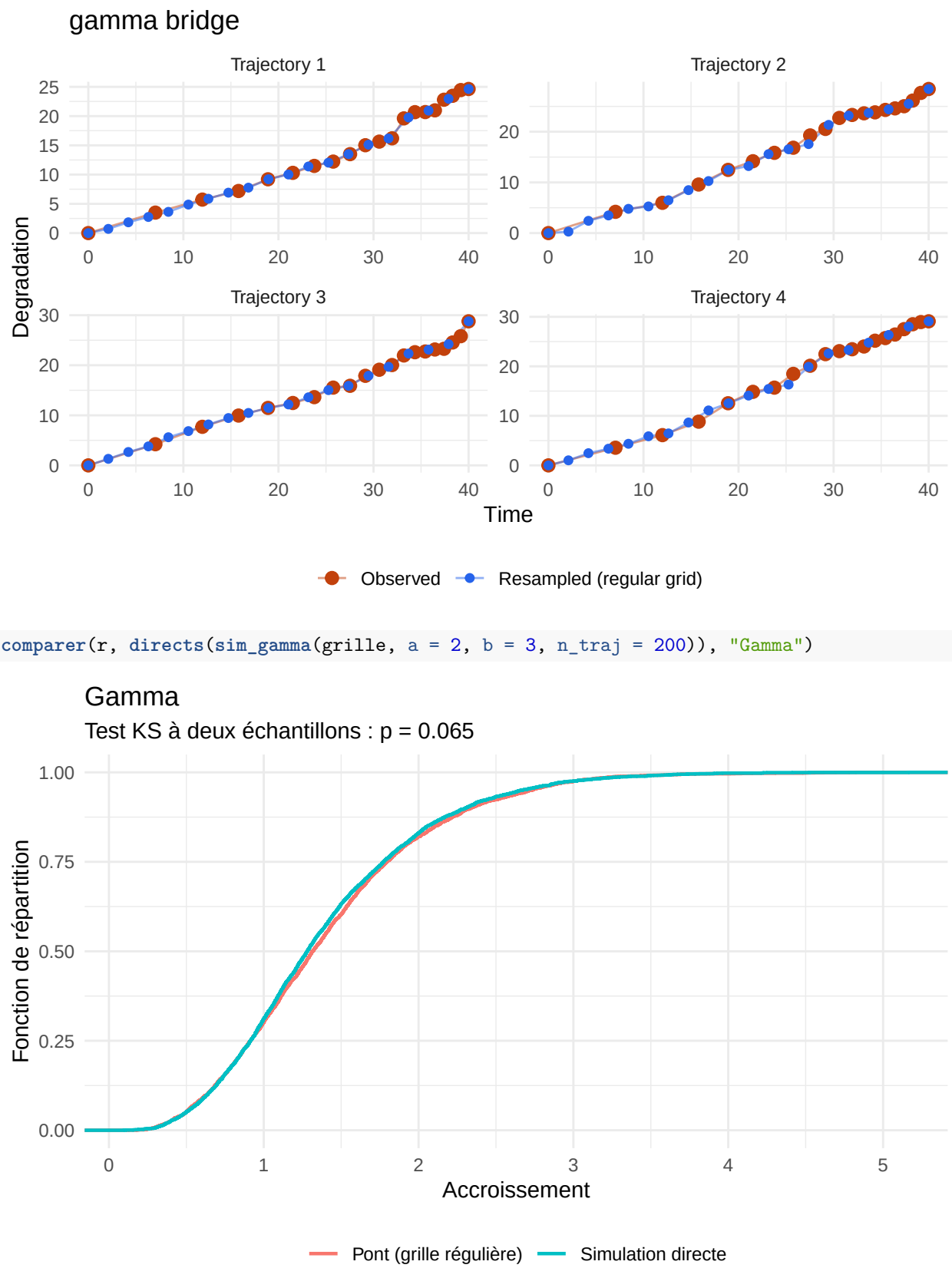
200 trajectoires, chacune inspectée 20 fois entre 0 et 40, de plus en plus souvent. On compare les accroissements reconstruits par le pont à des accroissements simulés directement sur la grille régulière : ils doivent avoir la même loi.

```
plan <- inspection_times("increasing", n = 19, n_traj = 200, tmax = 40)
grille <- seq(0, 40, length.out = 20)

comparer <- function(r, directs, titre) {
  d <- rbind(data.frame(dx = increments(r)$dx, origine = "Pont (grille régulière)"),
             data.frame(dx = directs, origine = "Simulation directe"))
  p <- ks.test(increments(r)$dx, directs)$p.value
  ggplot(d, aes(x = dx, colour = origine)) +
    stat_ecdf(linewidth = 0.8) +
    labs(title = titre,
         subtitle = paste("Test KS à deux échantillons : p =", round(p, 3)),
         x = "Accroissement", y = "Fonction de répartition", colour = NULL) +
    theme_minimal() +
    theme(legend.position = "bottom")
}
directs <- function(x) unlist(lapply(x, function(d) diff(d$X)))
```

Gamma

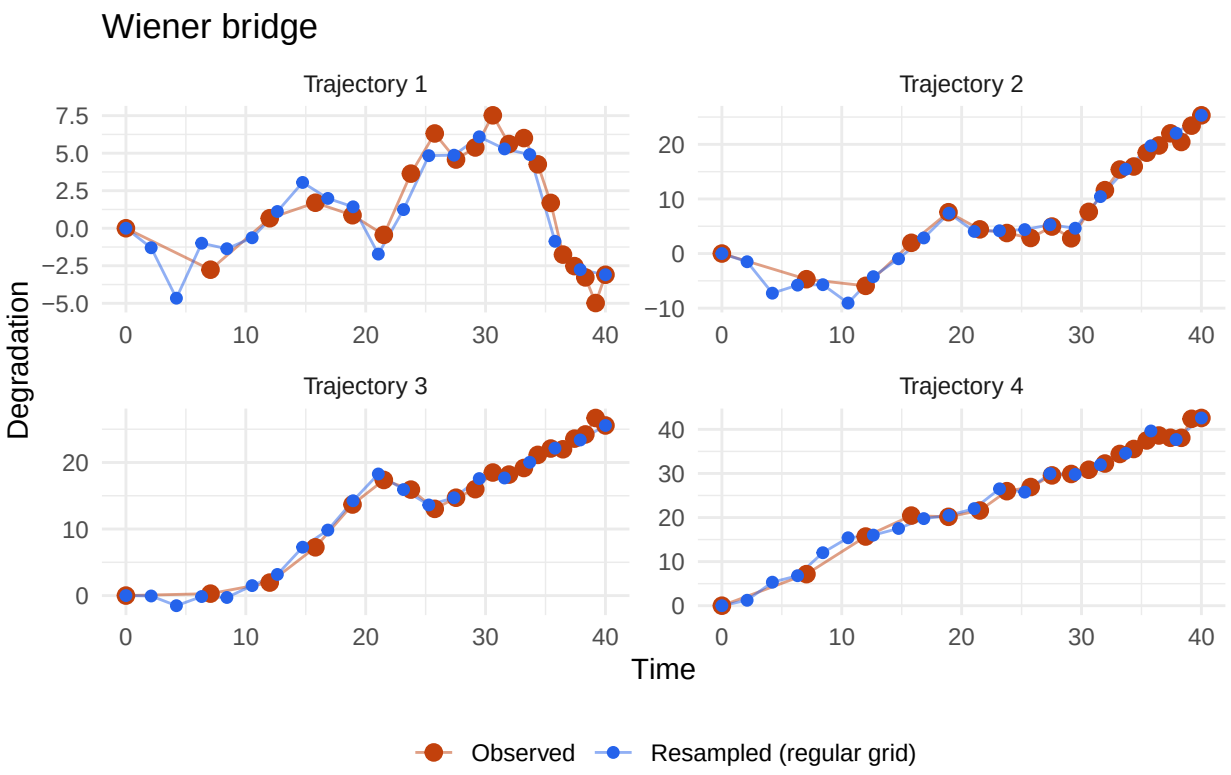
```
set.seed(2)
r <- gamma_bridge(sim_gamma(plan, a = 2, b = 3), a = 2)
plot_bridge(r, trajectory = 1:4)
```



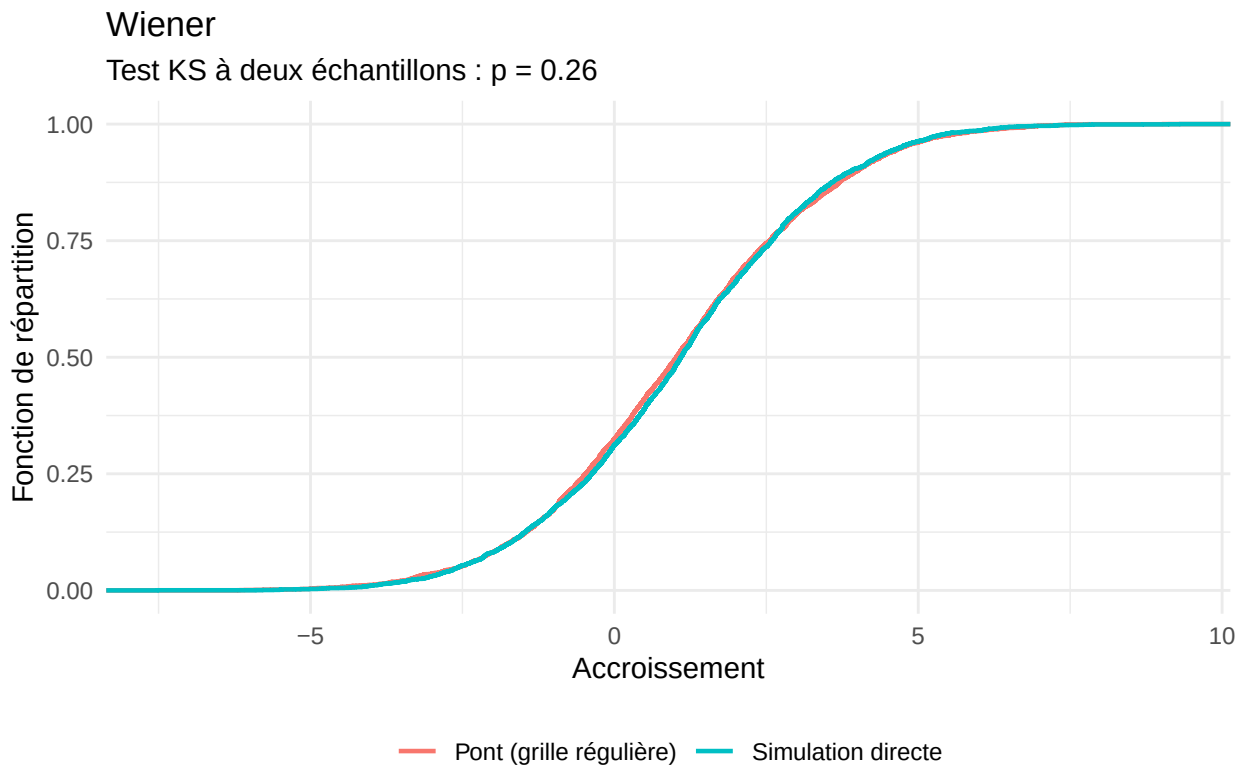

```
comparer(r, directs(sim_gamma(grille, a = 2, b = 3, n_traj = 200)), "Gamma")
```

Wiener

```
r <- wiener_bridge(sim_wiener(plan, mu = 0.5, sigma = 1.5), sigma = 1.5)
plot_bridge(r, trajectory = 1:4)
```

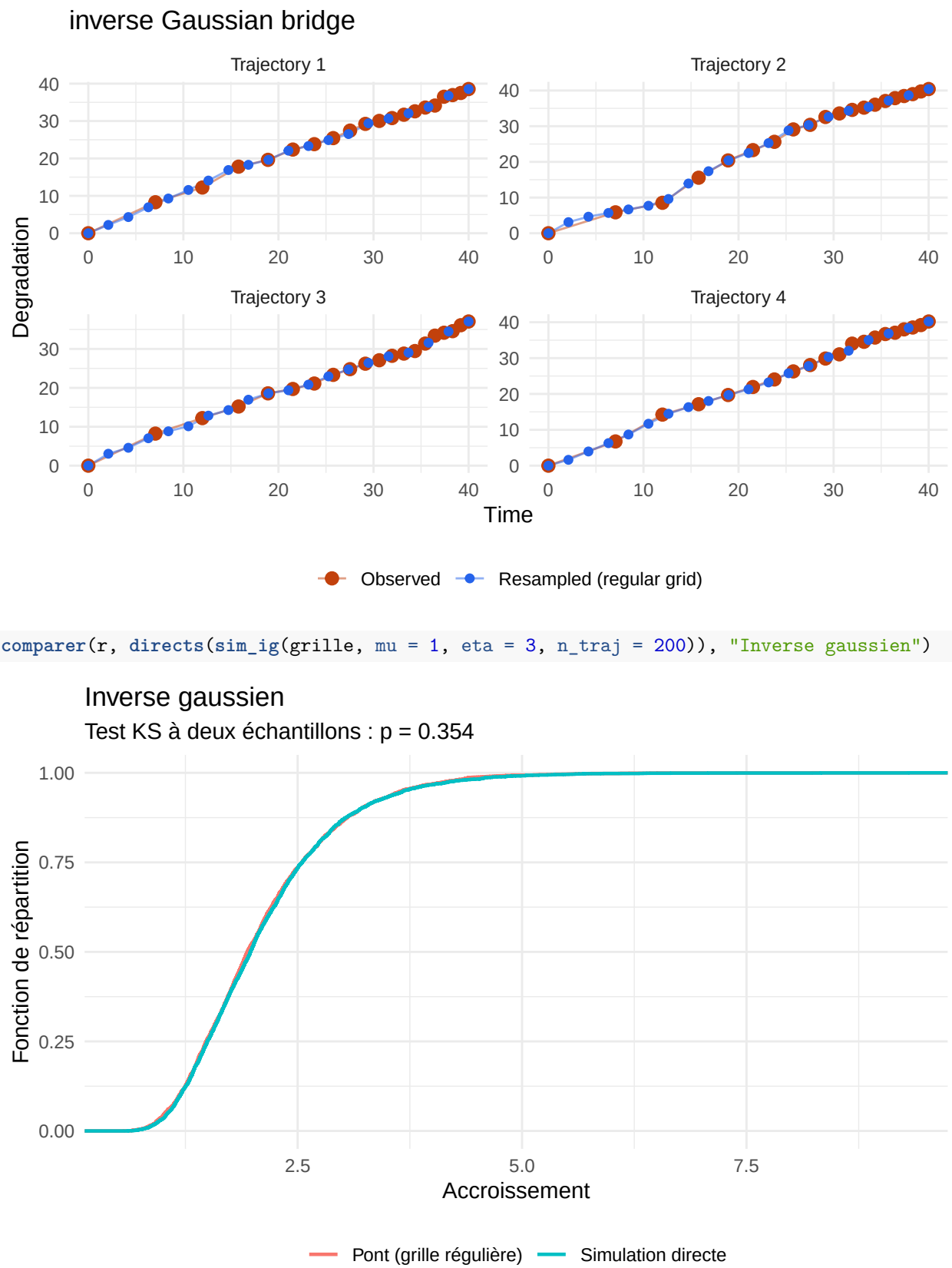



```
comparer(r, directs(sim_wiener(grille, mu = 0.5, sigma = 1.5, n_traj = 200)), "Wiener")
```



Inverse gaussien

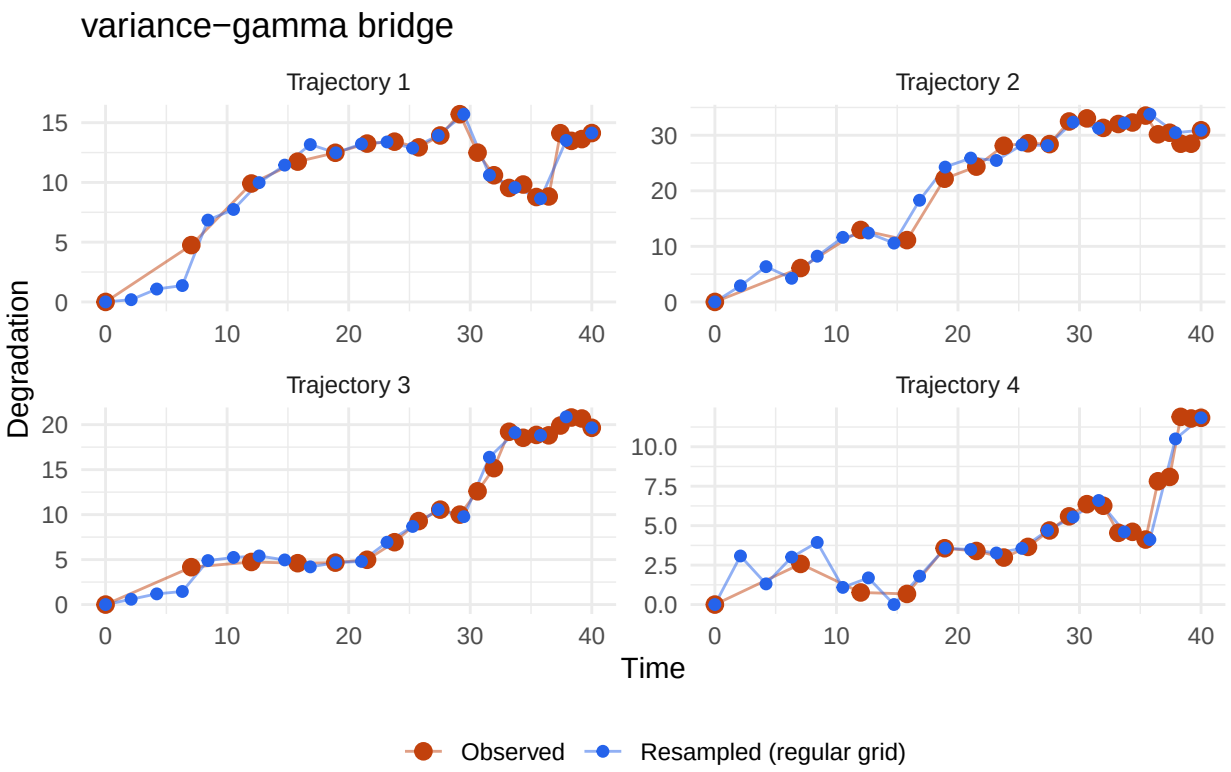
```
r <- ig_bridge(sim_ig(plan, mu = 1, eta = 3), eta = 3)
plot_bridge(r, trajectory = 1:4)
```



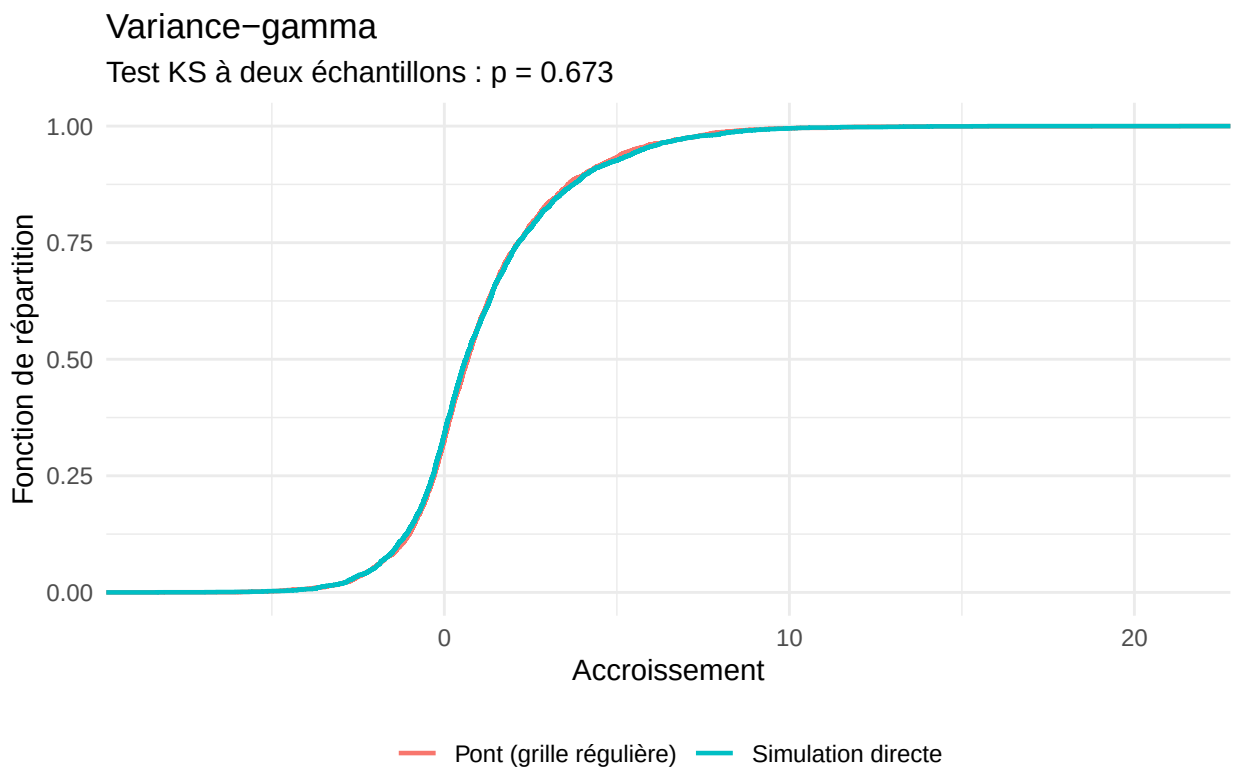

```
comparer(r, directs(sim_ig(grille, mu = 1, eta = 3, n_traj = 200)), "Inverse gaussien")
```

Variance-gamma

```
x <- sim_vg(plan, theta = 0.5, sigma = 1.5, nu = 2)
r <- vg_bridge(x, theta = 0.5, sigma = 1.5, nu = 2)
plot_bridge(r, trajectory = 1:4)
```

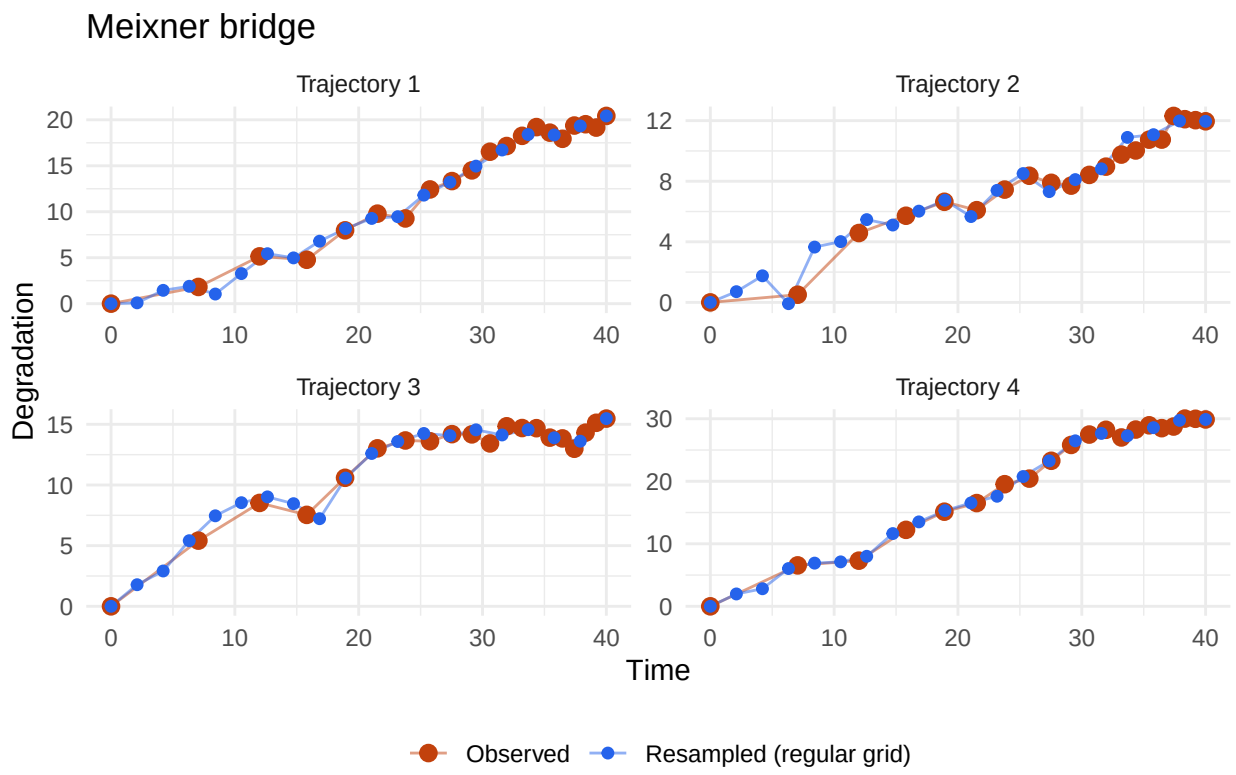



```
directs_vg <- directs(sim_vg(grille, theta = 0.5, sigma = 1.5, nu = 2, n_traj = 200))
comparer(r, directs_vg, "Variance-gamma")
```

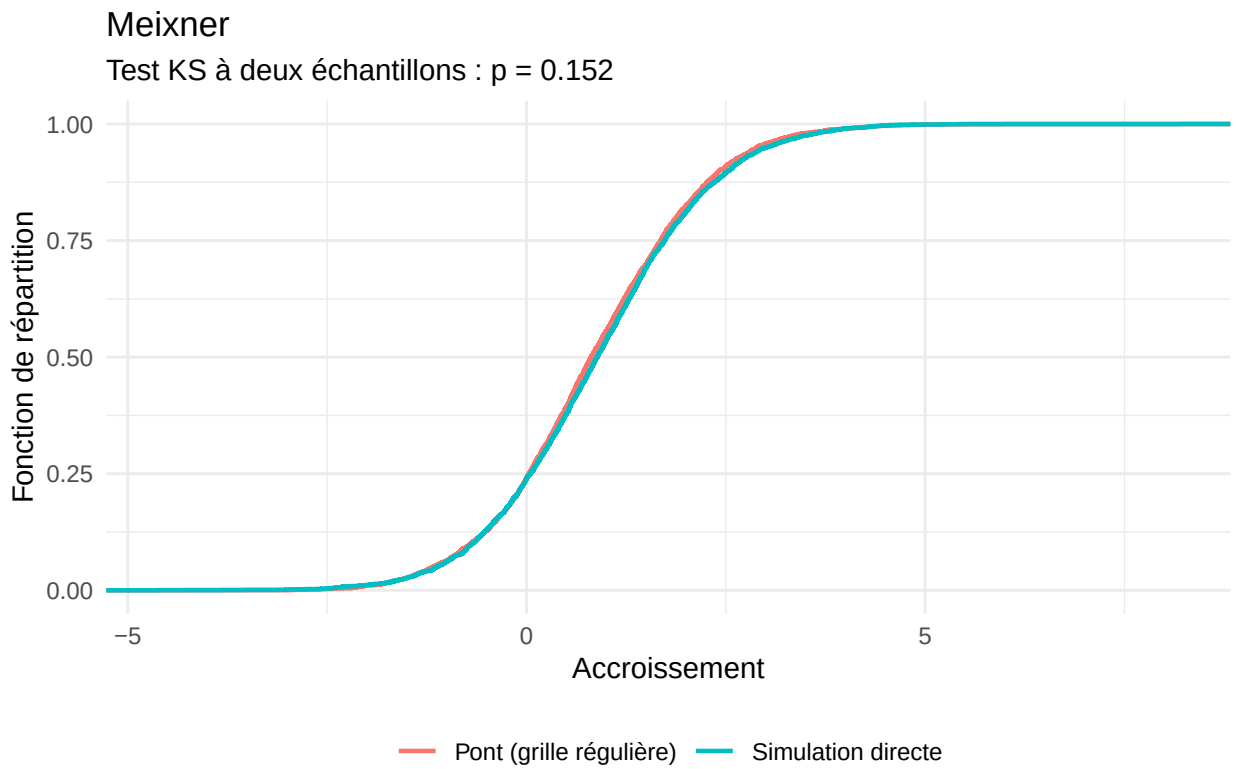


Meixner

```
x <- sim_meixner(plan, a = 1, b = 0.3, d = 1.5, m = 0.2)
r <- meixner_bridge(x, a = 1, d = 1.5, m = 0.2)
plot_bridge(r, trajectory = 1:4)
```




```
directs_mx <- directs(sim_meixner(grille, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 200))
comparer(r, directs_mx, "Meixner")
```



III - Données réelles : lasers GaAs

Les données `gaaslaser` donnent l'augmentation relative (en %) du courant de fonctionnement de 15 lasers, mesurée toutes les 250 heures jusqu'à 4000 heures. On ne garde que des inspections irrégulières, de moins en moins fréquentes, puis on estime le paramètre de forme du processus gamma avec `levyfit` et on reconstruit les trajectoires sur la grille régulière.

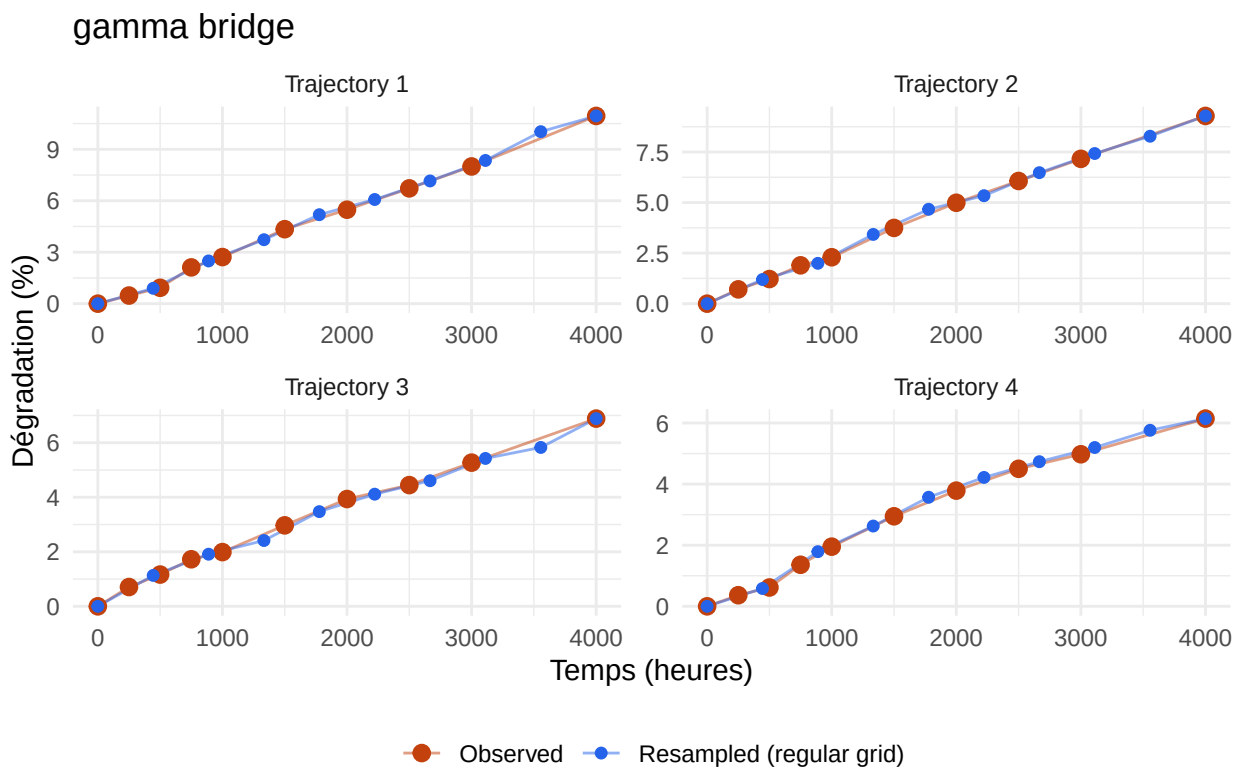
```
d <- SMRD.data::gaaslaser
temps <- c(0, 250, 500, 750, 1000, 1500, 2000, 2500, 3000, 4000)
lasers <- lapply(sort(unique(d$unit)), function(k) {
  s <- d[d$unit == k & d$hours %in% temps, ]
  data.frame(t = s$hours, X = s$increase)
})
est <- gamma_mle(lasers)
est
```

```
##           a           b
##  0.02166498 10.63099576
```

```
set.seed(3)
r <- gamma_bridge(lasers, a = est[["a"]])
r
```

```
## 15 trajectory(ies) resampled on a regular grid by the gamma bridge (method "Bridge")
## parameters: a = 0.02166
## time step teq: 444.4
##
##   id         t         X
## 1  1    0.0000 0.0000000
## 2  1  444.4444 0.8946652
## 3  1  888.8889 2.4910002
## 4  1 1333.3333 3.7287334
## 5  1 1777.7778 5.1872623
## 6  1 2222.2222 6.0730585
```

```
plot_bridge(r, trajectory = 1:4) + labs(x = "Temps (heures)", y = "Dégradation (%)")
```

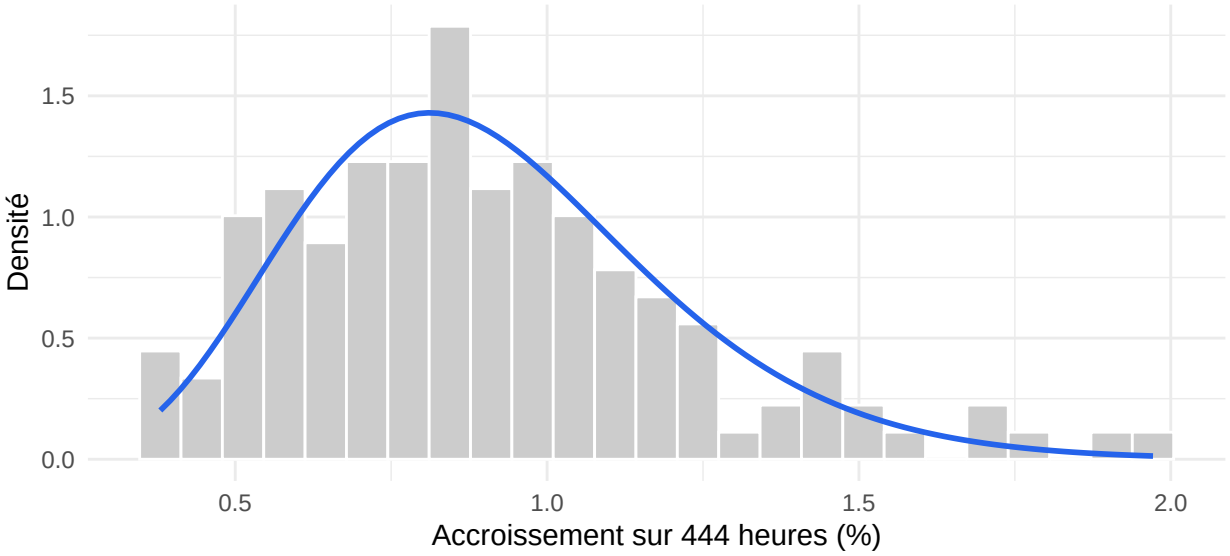


Accroissements reconstruits, sur une grille de pas $4000/9 \approx 444$ heures :

```
acc <- increments(r)
head(acc)
```

```
##   id      teq        dx
## 1  1 444.4444 0.8946652
## 2  1 444.4444 1.5963350
## 3  1 444.4444 1.2377331
## 4  1 444.4444 1.4585289
## 5  1 444.4444 0.8857962
## 6  1 444.4444 1.0817446
```

```
ggplot(acc, aes(x = dx)) +
  geom_histogram(aes(y = after_stat(density)), bins = 25, fill = "grey80", colour = "white") +
  stat_function(fun = dgamma, args = list(shape = est[["a"]] * acc$teq[1], rate = est[["b"]]),
                colour = "#2563EB", linewidth = 1) +
  labs(x = "Accroissement sur 444 heures (%)", y = "Densité") +
  theme_minimal()
```


Densité
1.5
1.0
0.5
0.0
0.5
1.0
1.5
2.0
Accroissement sur 444 heures (%)